Basic Meteorology

EES404 (4 Credits)

<u>Course Instructors</u>

Dr. Raju Attada & Dr. Ojha

Earth and Environmental Sciences

Office: AB2-5F5

Email: rajuattada@iisermohali.ac.in

Mobile: +91-9657192050

Meteorology

☞ The study of the atmosphere, which is associated with Weather and Climate.

☞ Deals the Physics, Dynamics and Chemical state of the earth's atmosphere.

Weather and Climate

☞ Weather is what you **see** ➡ The instantaneous state of the the atmospheric system.

☞ Climate is what you **expect** ➡ Long term mean of the atmospheric conditions, averaged over long period of time (over 30 years; for example).

WEATHER

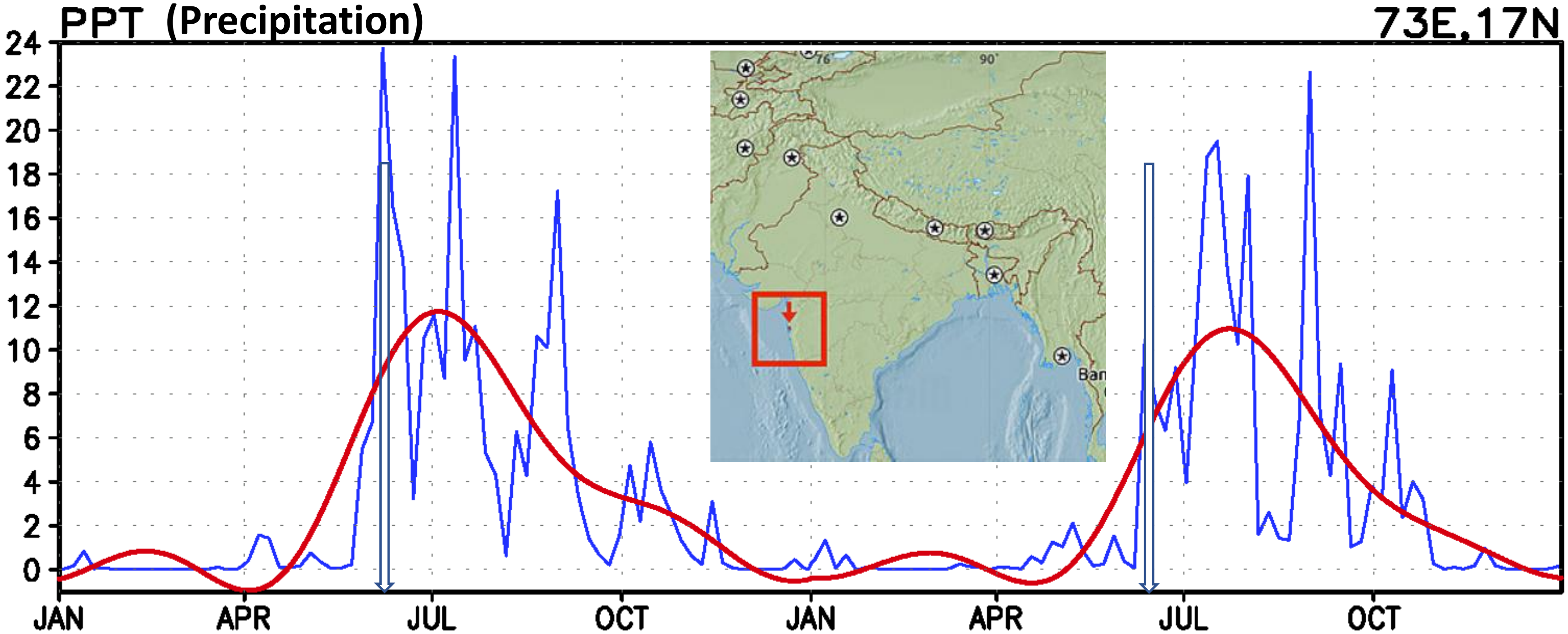
Tells you what to wear each day

CLIMATE

Tells you what types of clothes to have in your closet

https://www.ncei.noaa.gov/news/weather-vs-climate

An example of Weather and Climate:



Daily time series of "Year-2015 & 2016" precipitation (PPT) at Mumbai. The red line is the climate, blue line indicates weather.

- ✓ Weather can change within a short span of time (hours to days)
- ✓ Climate takes long time to change

➢ How do we describe the Weather or Climate ?

Temperature   Pressure   Winds   Humidity   Rainfall

➢ How do we observe the atmosphere ?

Ground based observational network (AWS)

Weather Balloons/Radiosonde network

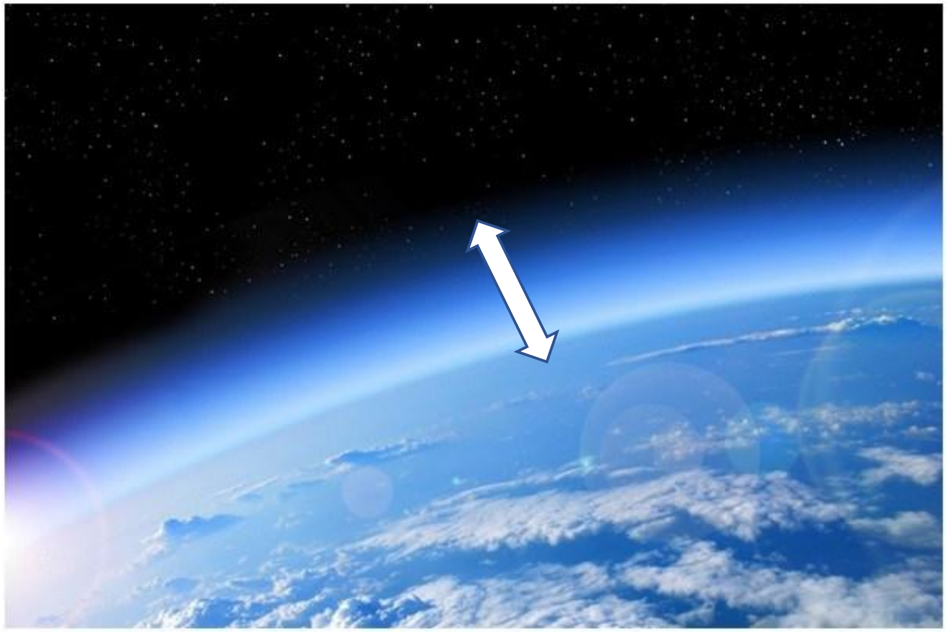
Space based platforms (Remote Sensing)

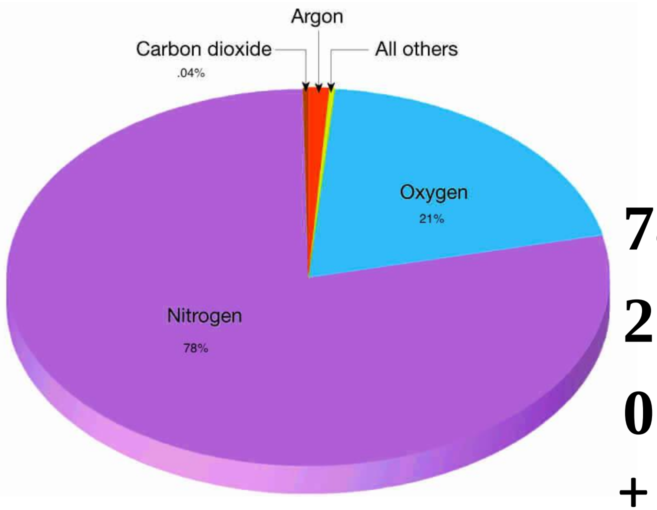
Weather Radars, Wind profilers

- Thermometer for measuring air and sea surface temperature
- Barometer for measuring atmospheric pressure
- Hygrometer for measuring humidity
- Anemometer for measuring wind speed
- Pyranometer for measuring solar radiation
- Rain gauge for measuring liquid precipitation over a set period of time.
- Wind sock for measuring general wind speed and wind direction
- Wind vane, also called a weather vane or a weathercock: it shows which way the wind is blowing.
- Present Weather/Precipitation Identification Sensor for identifying falling precipitation
- Disdrometer for measuring drop size distribution
- Transmissometer for measuring visibility
- Ceilometer for measuring cloud ceiling

Earth's Atmosphere

- A thin layer of the gaseous (air) surrounding the earth .
- A mixture of invisible gases (permanent and variable gases) and some microscopic particles.





78% Nitrogen (N_2)
21% Oxygen (O_2)
0.8% Argon (Ar)
+ Others

Majority of the atmosphere is occupied by N_2 and O_2

- **What is the importance of the atmosphere ?**

Composition of the Atmosphere

PERMANENT GASES		
Gas	Symbol	Percent (by Volume) Dry Air
Nitrogen	N_2	78.08
Oxygen	O_2	20.95
Argon	Ar	0.93
Neon	Ne	0.0018
Helium	He	0.0005
Hydrogen	H_2	0.00006
Xenon	Xe	0.000009

"Abundant Gases"

"Greenhouse Gases"

VARIABLE GASES			
Gas (and Particles)	Symbol	Percent (by Volume)	Parts per Million (ppm)*
Water vapor	H_2O	0 to 4	
Carbon dioxide	CO_2	0.038	380*
Methane	CH_4	0.00017	1.7
Nitrous oxide	N_2O	0.00003	0.3
Ozone	O_3	0.000004	0.04†
Particles (dust, soot, etc.)		0.000001	0.01–0.15
Chlorofluorocarbons (CFCs)		0.00000002	0.0002

Permanent gases

Nitrogen (78%):

- N_2 is added and removed from the atmosphere very slowly – long *residence time* of <u>~ 42 million years</u>.
- N_2 is relatively not **<u>important for meteorological and climate processes</u>**
- <u>Removal from: Biological process, ocean plankton & Adding from decaying of plant and animal matter</u>

Oxygen (21%):

- • O_2 is crucial to the existence of almost all forms of life on the Earth. Its *residence time* is <u>~5000 years.</u>
- <u>Removal: When organic matter decays & combines with other substances</u>

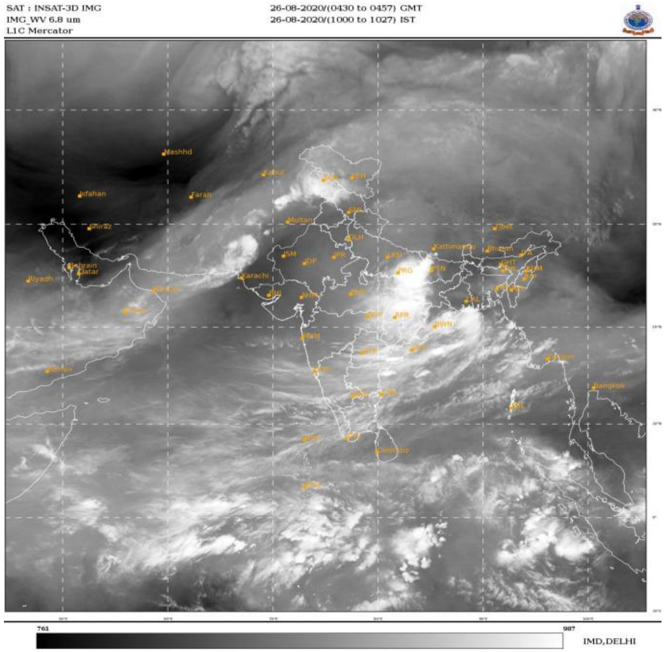
Variable gases

(distributions vary both in time and space)

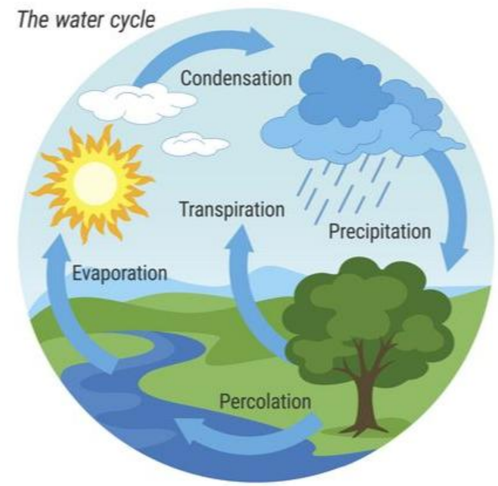
Constituent	Formula	Percent by Volume	Molecular Weight
Water Vapor	H_2O	0.25	18.01
Carbon Dioxide	CO_2	0.036	44.01
Ozone	O_3	0.01	48.00

- ☞ Account for < **1** % of the atmosphere
- ☞ These gases impact the behavior of the atmosphere

Water Vapour



- Water vapor is an extremely important gas in the atmosphere (e.g. cloud formation).
- Its highly variable in both space and time.
- Continually cycled between atmosphere and earth by evaporation, condensation and precipitation.
- Water vapor has a residence time of only **10 days**.



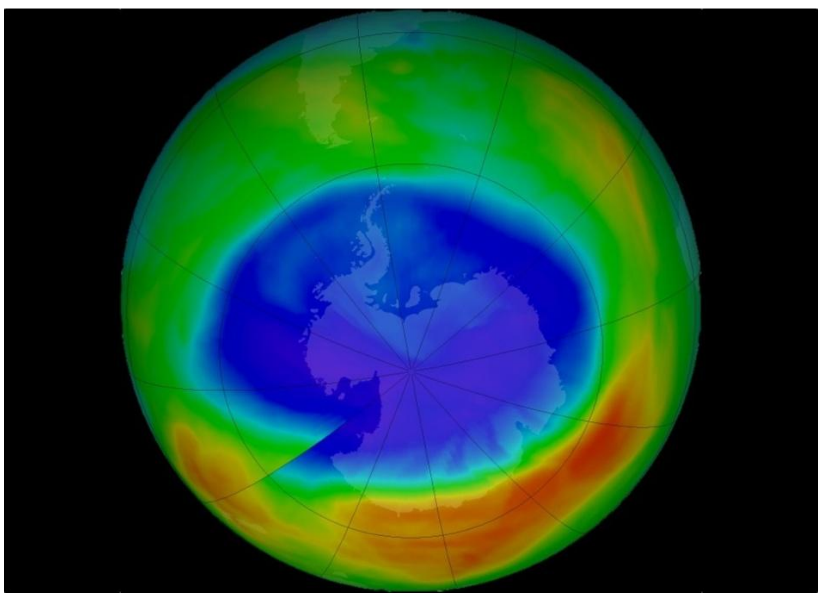
- WV **absorbs radiant energy emitted from the Earth's surface (GHG)**.

➢ **What is the uniqueness of the water vapour ?**

Ozone

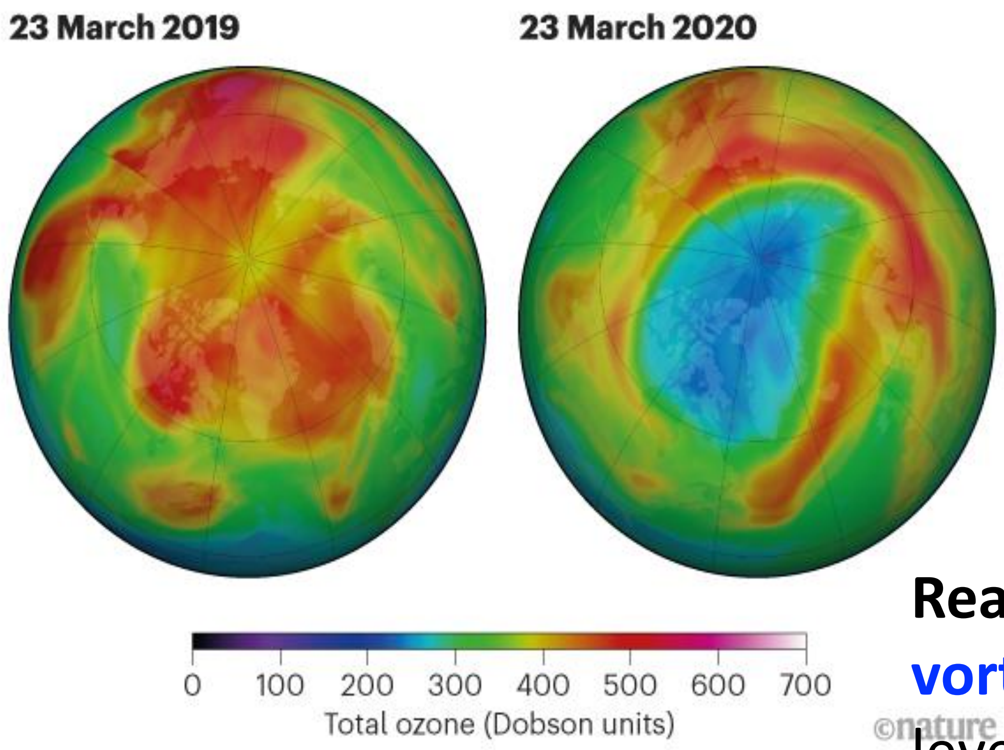
- ☞ O_3 is an unusual molecule made up of 3 Oxygen atoms. Majority of the atmospheric ozone is found in the upper troposphere/stratosphere.
- ☞ O_3 is essential element for absorbing harmful UV radiation from the sun.
- ☞ Near the surface ozone is a pollutant, and is the primary ingredient of photochemical smog (Smoke+Fog), which irritates the eyes and throat and damages vegetation.

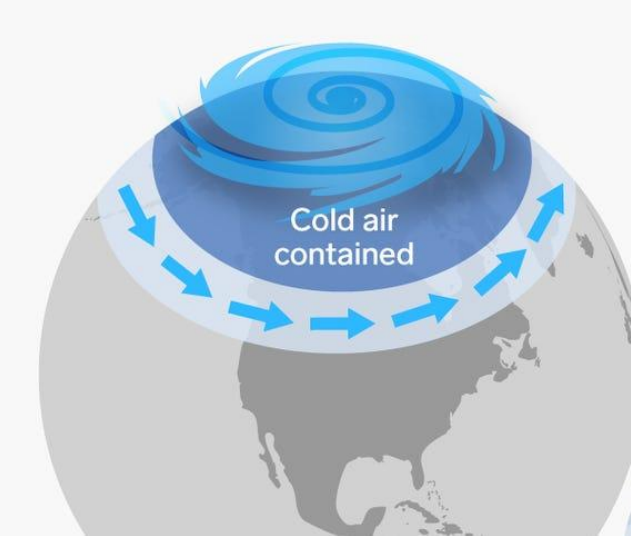
Satellite image showing depletion of ozone (ozone hole)



- ☞ Chlorofluorocarbons (CFCs) are believed to be depleting upper atmospheric ozone.
- ☞ The most prominent time for the depletion is **Winter**

➢ **What happens if ozone has depleted ?**



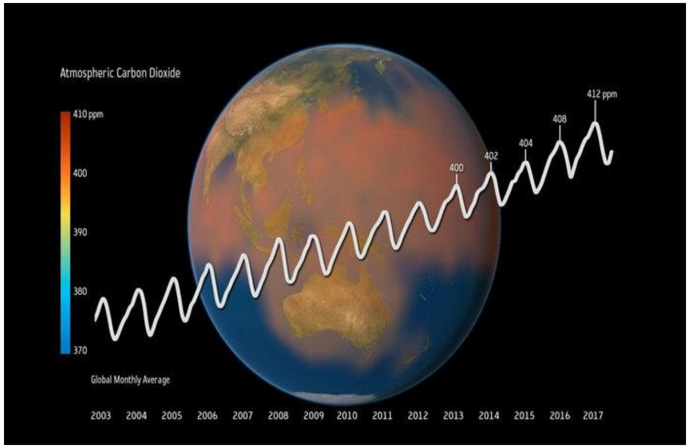


Reason: Presence of **polar vortex** (an upper-level/stratosphere low-pressure area), cold temperatures (-80 C) and CFCs

☞ **Potential way to mend the ozone hole is: Stop releasing CFCs and other ozone depleting gases into the atmosphere.**

☞ **Arctic ozone loss tends to be far less severe than in the Antarctic**

Carbon Dioxide



- ☞ CO_2 is supplied to the atmosphere through **<u>decay of vegetation, plant and animal respiration, volcanic eruptions, from the burning of fossil fuels (such as coal, oil, and natural gas), and from deforestation.</u>**

- ☞ CO_2 has a residence time of <u>>10 years</u>.

- ☞ It is removed through **<u>photosynthesis</u>**, "as plants consume CO2 to produce green matter".

- ☞ The oceans act as a huge reservoir for CO2

- ☞ Carbon dioxide is another important greenhouse gas because, like water vapor, it traps a portion of the earth's outgoing energy (GHG).

- ☞ **Measurements indicate that the concentration in the atmosphere has increased ~30% since 1958.**

It means "CO2 is entering the atmosphere at a greater rate than it is being removed"

June 2025: 429.61 ppm

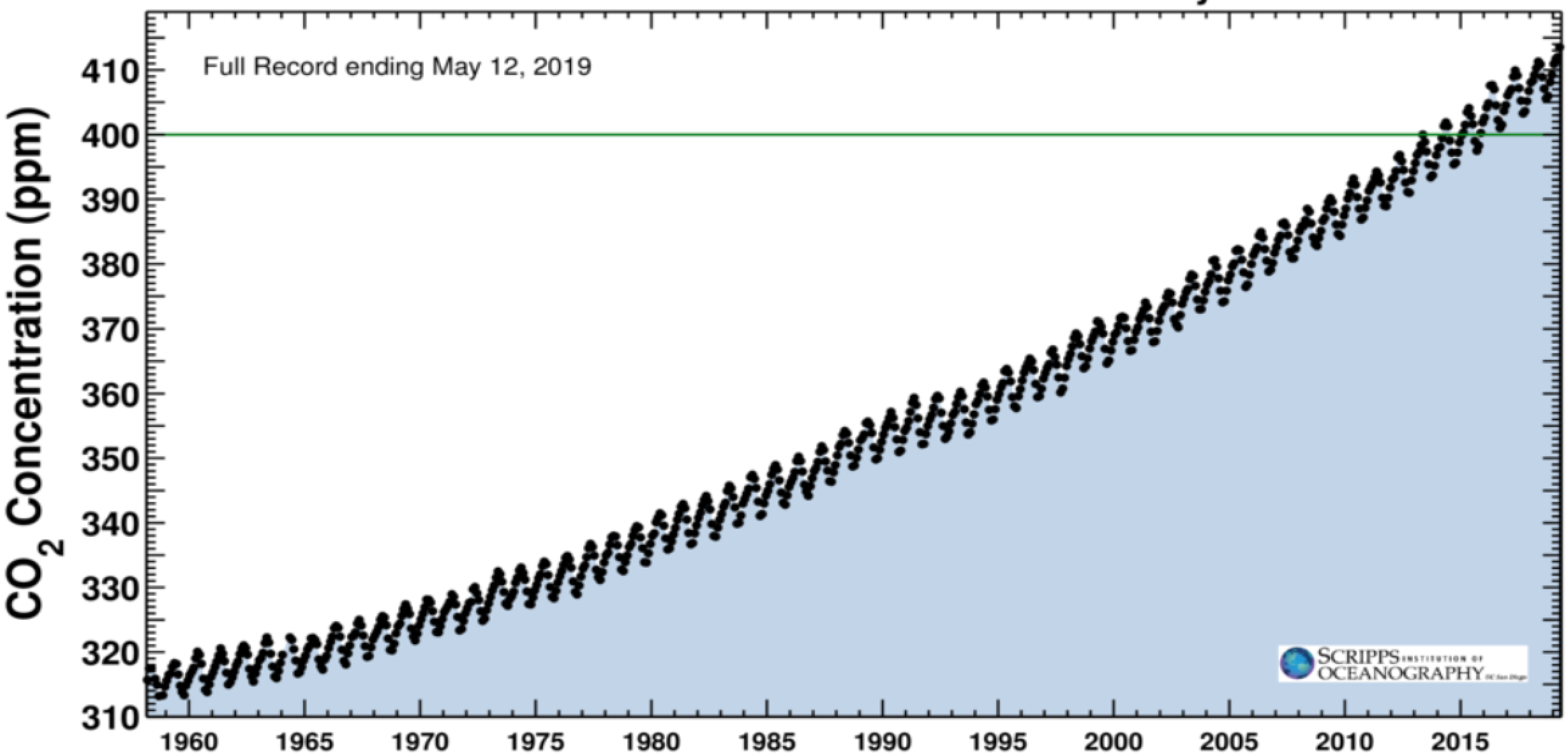
June 2024: 426.91 ppm

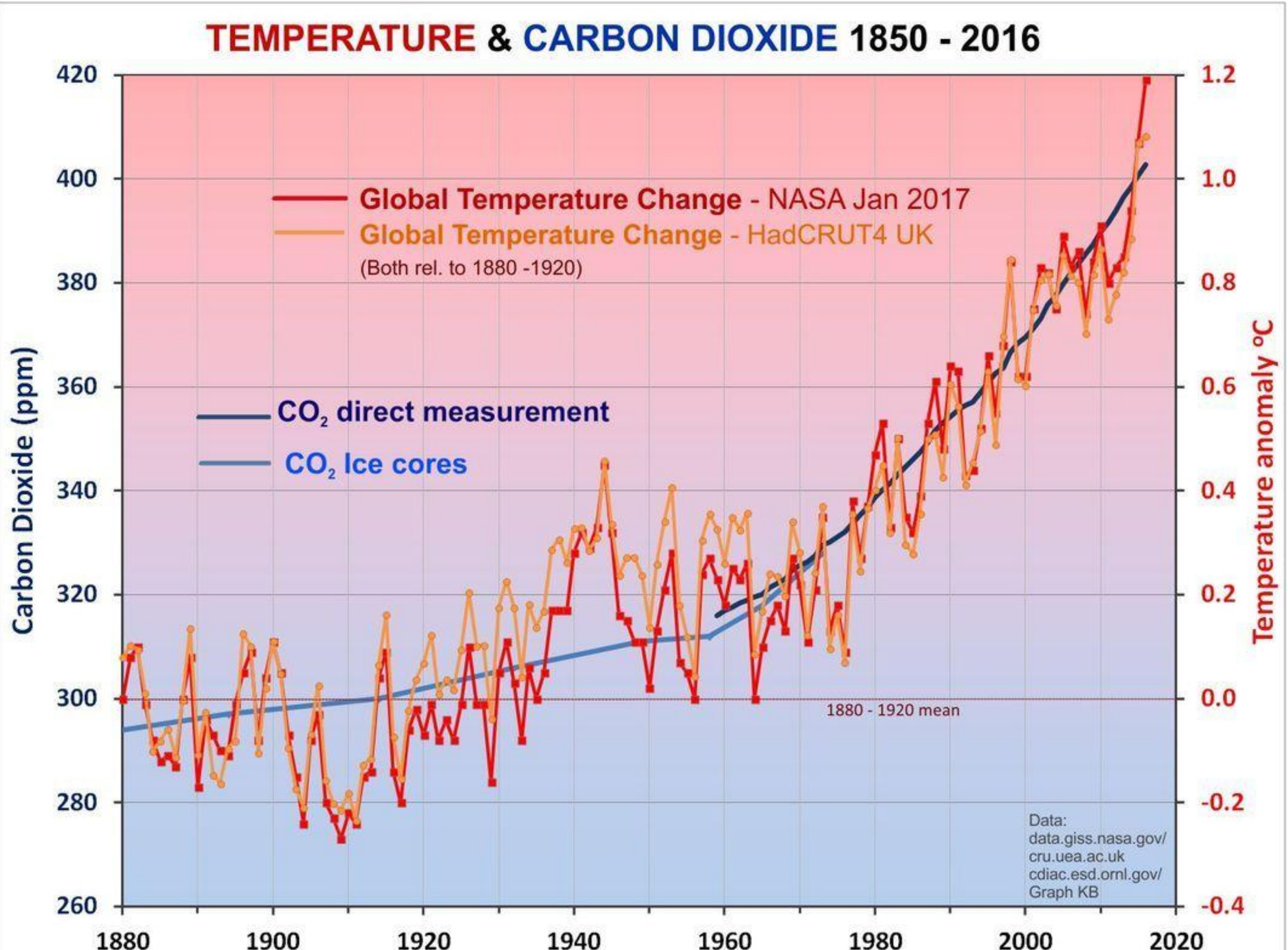
Last updated: Jul 14, 2025

Latest CO_2 reading
May 12, 2019

415.39 ppm

Carbon dioxide concentration at Mauna Loa Observatory





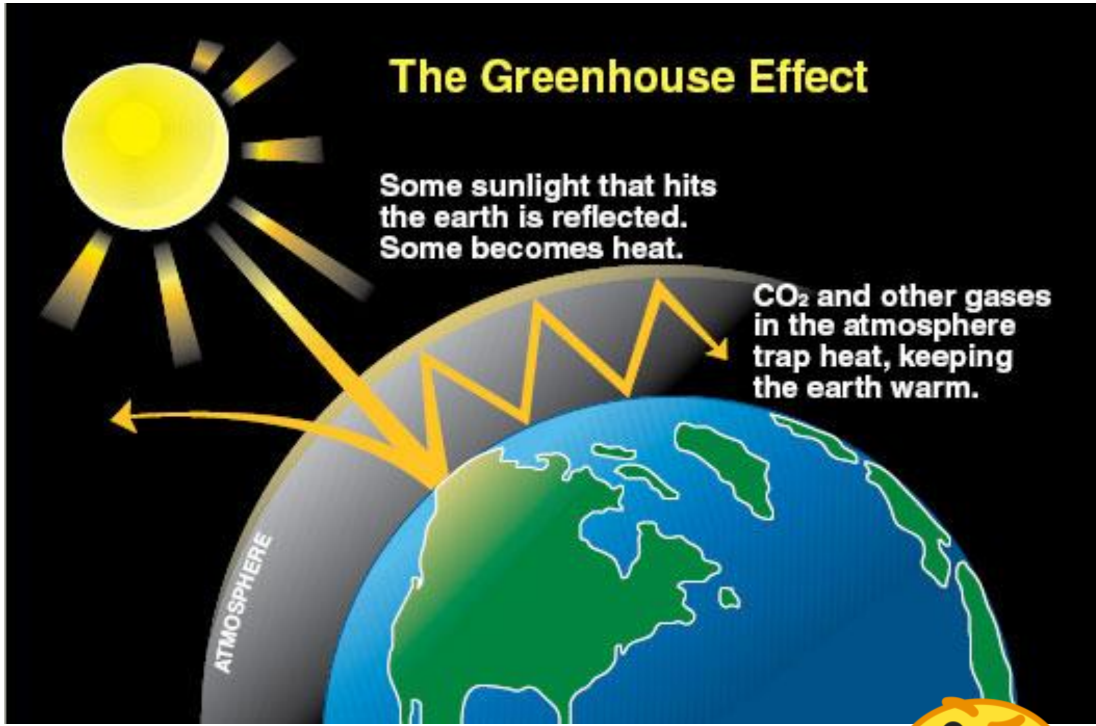
TEMPERATURE & CARBON DIOXIDE 1850 - 2016
Global Temperature Change - NASA Jan 2017
Global Temperature Change - HadCRUT4 UK
(Both rel. to 1880 -1920)
CO_2 direct measurement
CO_2 Ice cores
1880 - 1920 mean
Data:
data.giss.nasa.gov/
cru.uea.ac.uk
cdiac.esd.ornl.gov/
Graph KB
Carbon Dioxide (ppm)
Temperature anomaly °C
420
400
380
360
340
320
300
280
260
1.2
1.0
0.8
0.6
0.4
0.2
0.0
-0.2
-0.4
1880
1900
1920
1940
1960
1980
2000
2020

What is a Greenhouse Gas ?

- GHG is a gas that relatively transparent to solar radiation, but absorbs outgoing longwave radiation from the earth.
- These GHGs traps the heat in the atmosphere.

Predominant GHGs:

- Water vapor
- Carbon dioxide
- Ozone
- Nitrous oxide
- Methane



➢ **What are the consequences of Global Warming?**

Energy receive by the Earth ?

Short-wave radiation (SW): < --- Incoming

- As the sun's radiant energy travels through space, nothing interferes with it until it reaches the atmosphere.
- $\lambda < 4\ \mu m$ -- Energy associated with the solar radiation

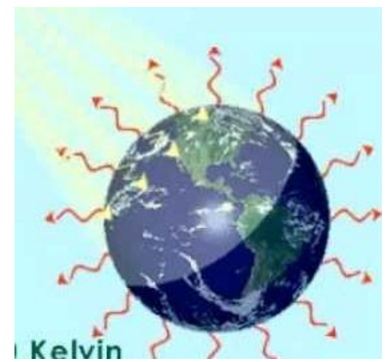
Solar Constant

Solar energy received on a surface perpendicular to the sun's rays appears to remain constant,

$1367\ W/m^2$ — Solar constant

Long-wave Radiation (LW): Outgoing --- >

- Outgoing Long-wave Radiation is electromagnetic radiation of wavelengths between 4.0 and 100 μm emitted from Earth and its atmosphere

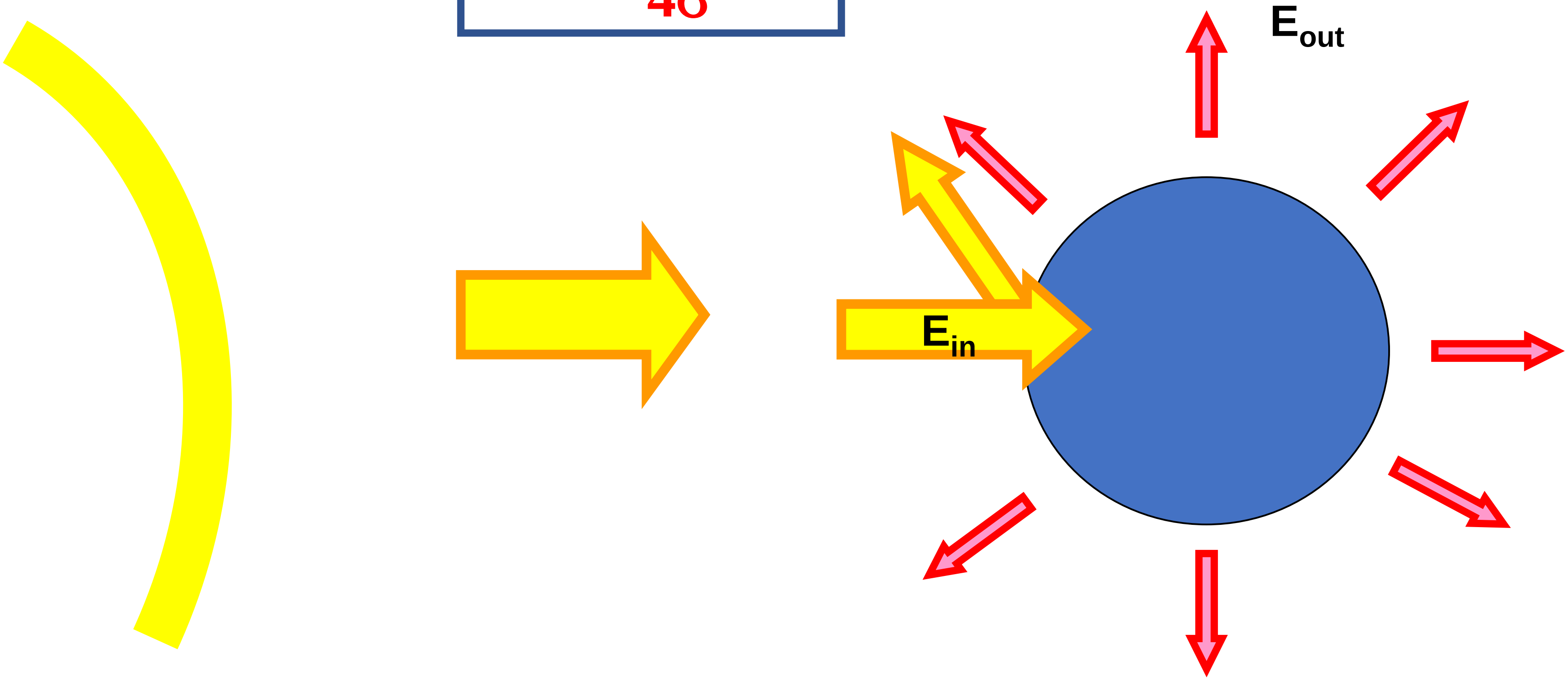


Effective temperature of the Earth surface:

$$E_{in} = E_{out}$$

$$S_o\,(1\text{-}A) = \sigma\, T^4\,(4)$$

$$\boxed{T^4 = \frac{S_o(1\text{-}A)}{4\sigma}}$$



$$T^4 = \frac{S_o(1-A)}{4\sigma}$$

If we know S_o and A, we can calculate the temperature of the Earth. We call this the expected temperature (T_{exp}). It is the temperature we would expect if Earth behaves like a blackbody.

This calculation can be done for any planet, provided we know its solar constant and albedo.

Calculation of Radiative Equilibrium Temperature

LW radiation lost = Incoming solar radiation

$$\tau 4\pi R^2 \sigma T_e^4 = \pi R^2 (1-\alpha) S_0$$

Solar constant $S_0 \approx 1365$ W m^{-2}

Albedo $\alpha = 0.3$

$\boldsymbol{T_e}$ – Radiative equilibrium temperature

τ – Infrared transmissivity (assuming no atmosphere, $\tau = 1.0$)

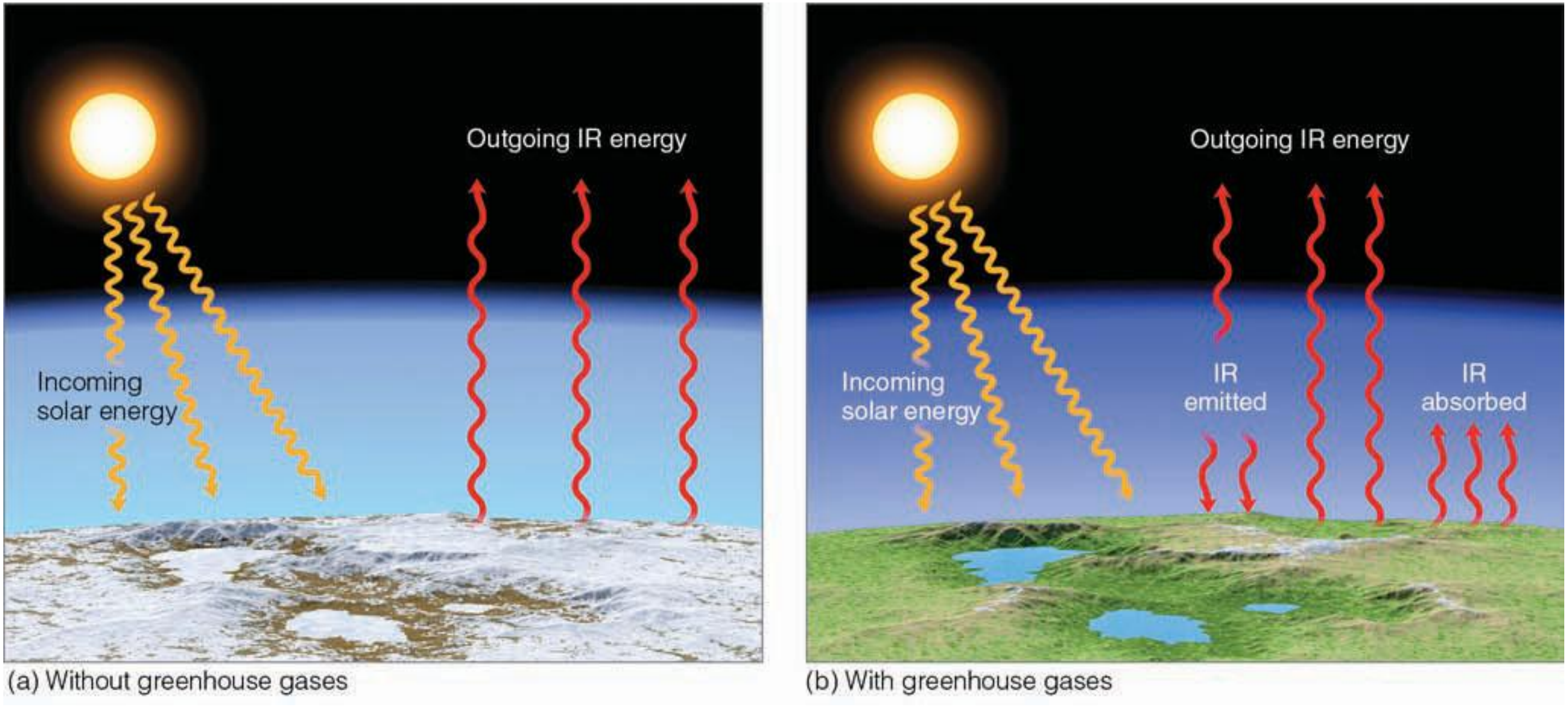
$$\sigma T_e^4 = \tfrac{1}{4}(1-\alpha) S_0 = 238 \text{ W m}^{-2}$$

or $\boxed{T_e = 255 \text{ K } (-18\,^\circ\text{C})}$ where $\sigma = 567 \times 10^{-8}$ W m^{-2} K^{-4}

Our current global mean surface temperature $T_{sfc} \approx 288$ K

ΔT = (288 - 255) = 33 K is the natural greenhouse effect of the earth's atmosphere.

Greenhouse effect: Good or Bad ?



- **The ground in the right picture must be warmer to be able to emit 3 arrows of ENRGY.**
- **Remember that the amount of energy emitted by something depends on temperature.**

The advantages of greenhouse gases

- The greenhouse effect helps to maintain a certain temperature level on Earth's surface, making it habitable for the living beings (humans, animals, plants and other organisms). Due to presence of theses greenhouse gases, the earth is warm enough to sustain life. Otherwise, the heat would have escaped the atmosphere, making earth's surface way cooler than what it is.

- The greenhouse gases help block the harmful solar radiation from reaching planet's surface. These gases work like a filter and bounce back most of the unwanted and damaging energy into space.

- Ozone, which is one of the crucial greenhouse gases, absorbs the harmful ultra-violet (UV) rays of the sun. If there wasn't any Ozone layer in our atmosphere, the UV rays could reach the surface directly and affect the life on earth on a massive scale.

- The greenhouse effect has allowed the planet to maintain its water level on the surface. Thanks to the moderate temperature, the planet's ice has not been melted completely, and the polar ice caps are restricted to the polar regions of the planet.

Equilibrium Temperature for Venus

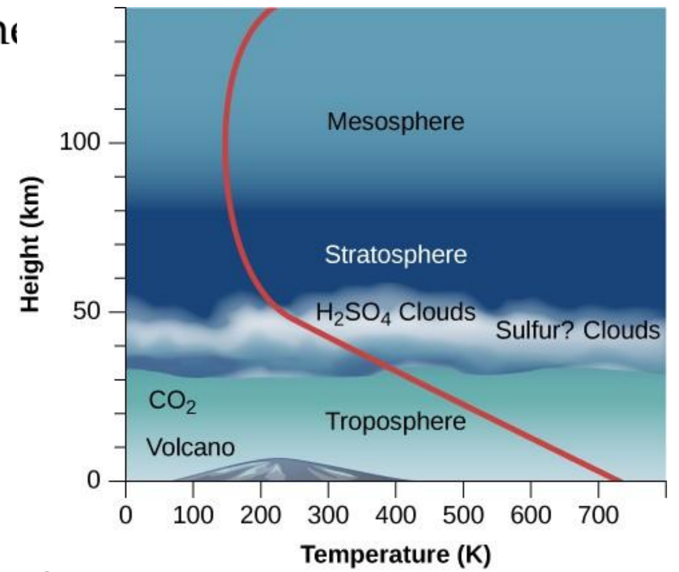
For Venus, $S_0 = 2643$ W m^2. It is almost covered with thick CO_2 clouds. Therefore $\alpha =$ 0.77.

If the Venus atmosphere were transparent to LW radiation, the equilibrium temperature would be

$$\sigma T_e^4 = \frac{(1-\alpha)S_0}{4} \approx 277 \text{ K}$$

However, the surface temperature is 750 K. Thus the

atmosphere is 750 - 277 = 473 K !



Because Co_2 on Venus is 96% and 4% N_2

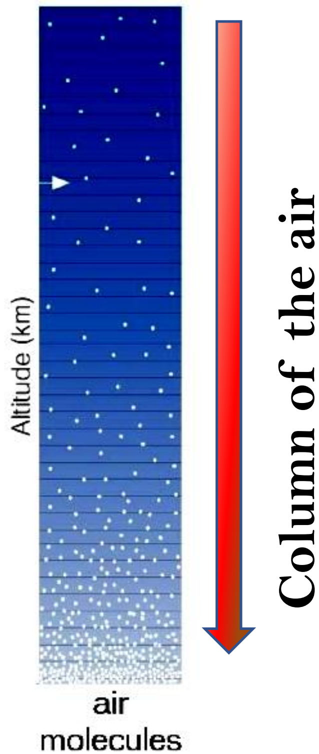
Chemical composition:

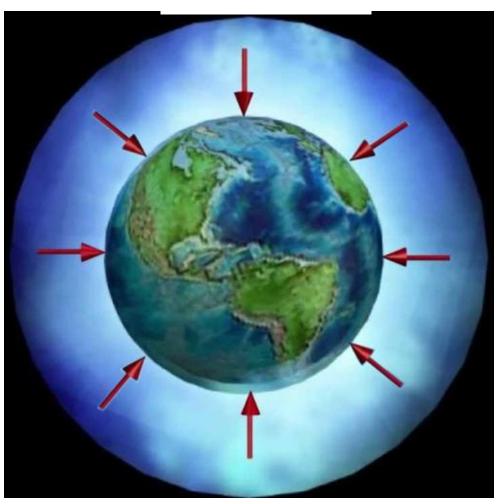
- ❑ **Carbon dioxide: 96 %**
- ❑ **Nitrogen: 3.5 %**
- ❑ **Carbon monoxide, Argon, sulfur dioxide, and water vapor: 0.5 percent**

Vertical Structure of the atmosphere

- Density
- Pressure
- Temperature
- Relative humidity

Density of the atmosphere:





☞ **Density (ρ) is the mass of air molecules per unit volume.** $\rho = m / V$

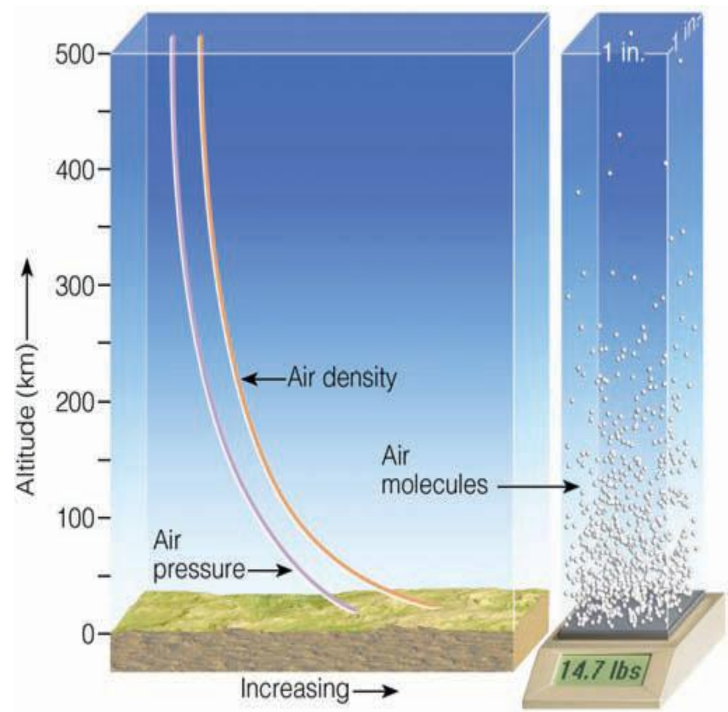
☞ **Most of air the molecules are held at the surface due to gravity.**

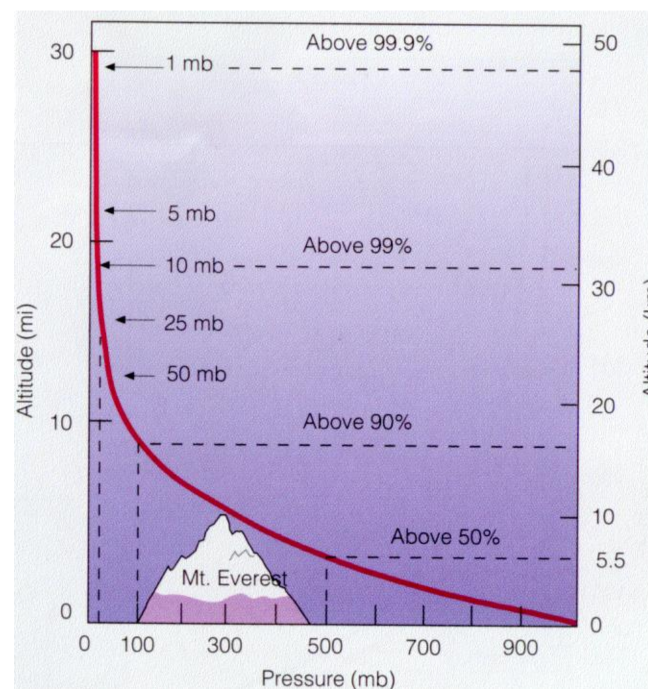
☞ **Density of the air molecules <u>decreases</u> with the height.**

☞ **Density at the mean sea level: 1.23 Kg/m^3**

Atmospheric Pressure

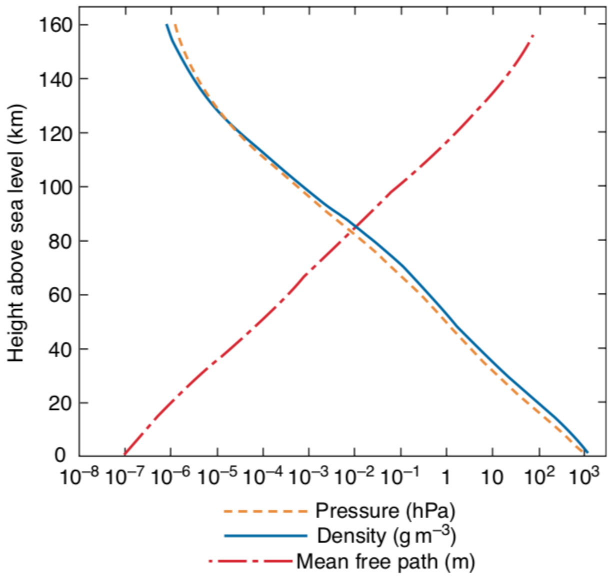
- ☞ The amount of force exerted over an area of surface is called atmospheric pressure
- ☞ The pressure at any level in the atmosphere may be measured in terms of <u>the total mass of air above any point</u>
- ☞ Atmospheric pressure always <u>decreases</u> with increasing height.
- ☞ Atmospheric Pressure at mean sea level: <u>1013.25 hPa </u>(milli bar).
- ☞ The pressure at any level (height) in the atmosphere measured in terms of the total mass of the air above.





The density of air at sea level is 1.25 kg/m^3. Pressure p and density ρ decrease nearly exponentially with height, i.e.,

$$p \simeq p_0 e^{-z/H}$$



where H, the e-folding depth, is referred to as the ***scale height*** and p_0 is the pressure at some reference level, which is usually taken as sea level (z =0).

The height within which pressure or density, decreases by a factor 1/e

In the lowest 100 km of the atmosphere, the scale height ranges roughly from 7 to 8 km.

Dividing Eq. by P_0 and taking the natural logarithms:

$$\boxed{\ln \frac{p}{p_0} \simeq -\frac{z}{H}}$$

☞ **This relationship is useful for estimating the height of various pressure levels in the Earth's atmosphere.**

Mass of the atmosphere:

At any point on the Earth's surface, the atmosphere exerts a downward force on the underlying surface due to the Earth's gravitational force.

The downward force, (i.e., the *weight*) of a unit volume of air with density ρ is given by

$$F = \rho \times g$$

Integrating from the Earth's surface to the top of the atmosphere,

we obtain the atmospheric pressure on the Earth's surface P_s due to the weight of the air in the overlying column:

$$P_s = \int_0^{\infty} \rho g \, dz$$

At the surface $g=g_0$ (9.81 ms^{-1})

$$P_s = g_0 \int_0^{\infty} \rho \, dz$$

$$\boxed{P_s = m \, g_0}$$

$$m = \int_0^{\infty} \rho \, dz$$

The vertically integrated mass per unit area of the overlying air.

The globally averaged surface pressure is 985 hPa. Estimate the mass of the atmosphere.

$P_s = 985 \times 10^2$ Pa ← Surface Pressure

$g = 9.81$ m/s^2 ← Accelaration due to gravity

$P_s = 985 \times 10^2$ $\cancel{9.81}$

$P_s = m \times g_0$

$\bar{m} = \dfrac{P_s}{g_0} = \dfrac{9.85 \times 10^4 \text{ Pa}}{9.81 \text{ m/sec}^2}$

$m = \boxed{1.004 \times 10^4 \text{ Kg/m}^2}$

Total mass of the atmosphere:

Assume the earth is sphere

Area of the sphere

6.37×10^6 mts

$4\pi R^2_{earth}$

R – radius of the earth

$$M_{earth} = m \times 4\pi R^2_{earth} \quad \text{(area of the earth)}$$

$$= 1.004 \times 10^4 \times \frac{Kg}{m^2} \times 4 \times 3.14 \times (6.37 \times 10^6)^2 \times m^2$$

$$= 5.10 \times 10^{18}\ Kg$$

This is amount of air above us.

At approximately what height above sea level z_m does half the mass of the atmosphere lie above and the other half lie below?

Hint: Assume an exponential pressure dependence with H = 8 km

Let P_m be the pressure level that half the mass of the atmosphere lies above and half lies below.

Hence $P_m = P_0/2$

$$\boxed{\ln \frac{p}{p_0} \simeq -\frac{z}{H}}$$

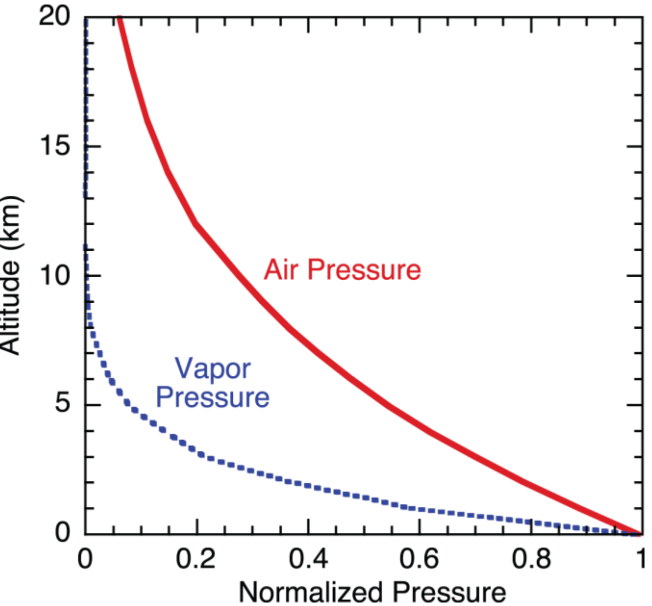
Substituting H = 8 km, we obtain

Z_m =8 Km × 0.693 ~ **5.5 km**

First 5.5 km, half the mass of the atmosphere $(\frac{1}{2}(5.1 \times 10^{18}$ Kg$))$ ==== $\mathbf{2.6 \times 10^{18}}$ **Kg**

Above 5.5 km height rest of the mass lie ===== $\mathbf{2.6 \times 10^{18}}$ **Kg**

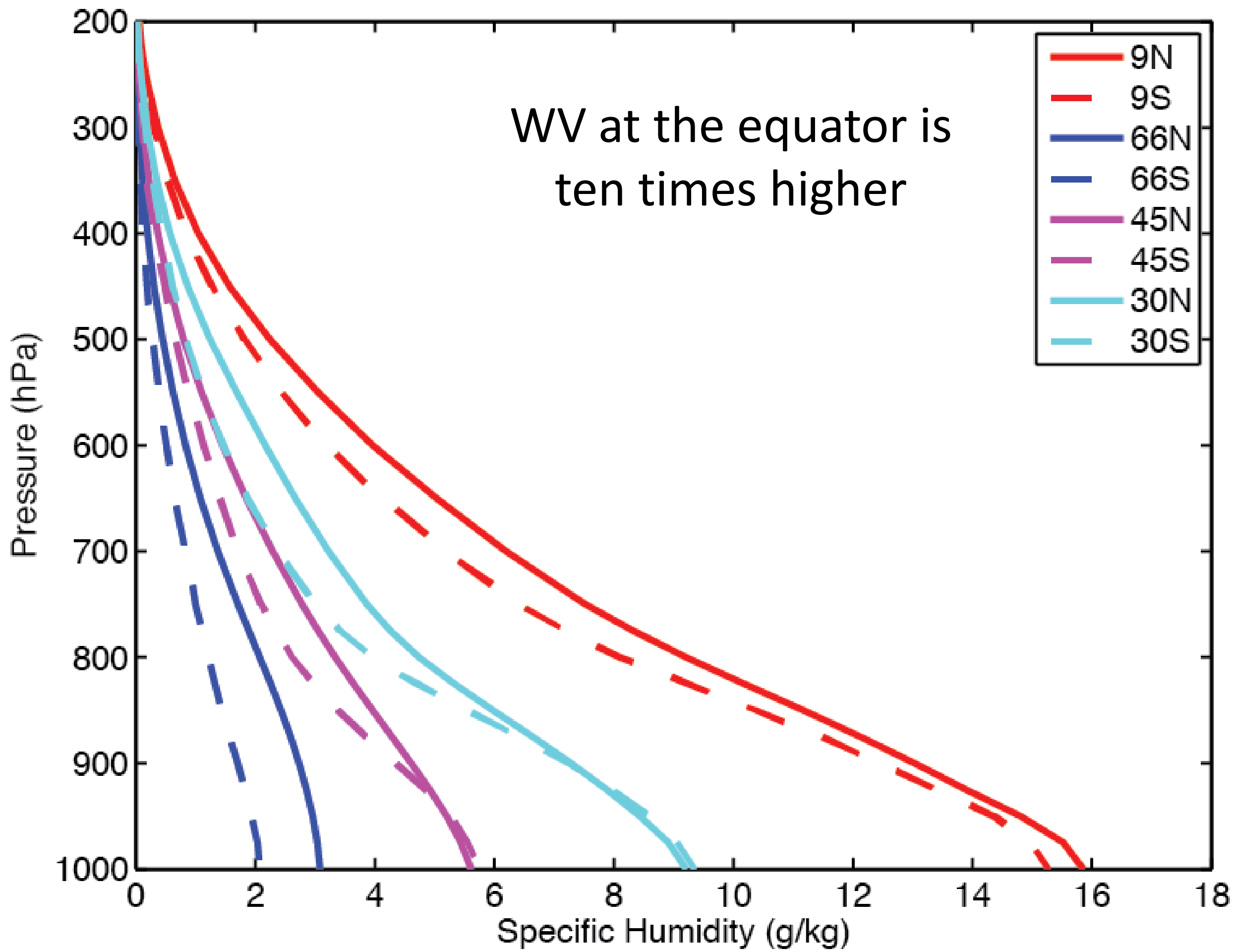
Atmospheric Humidity



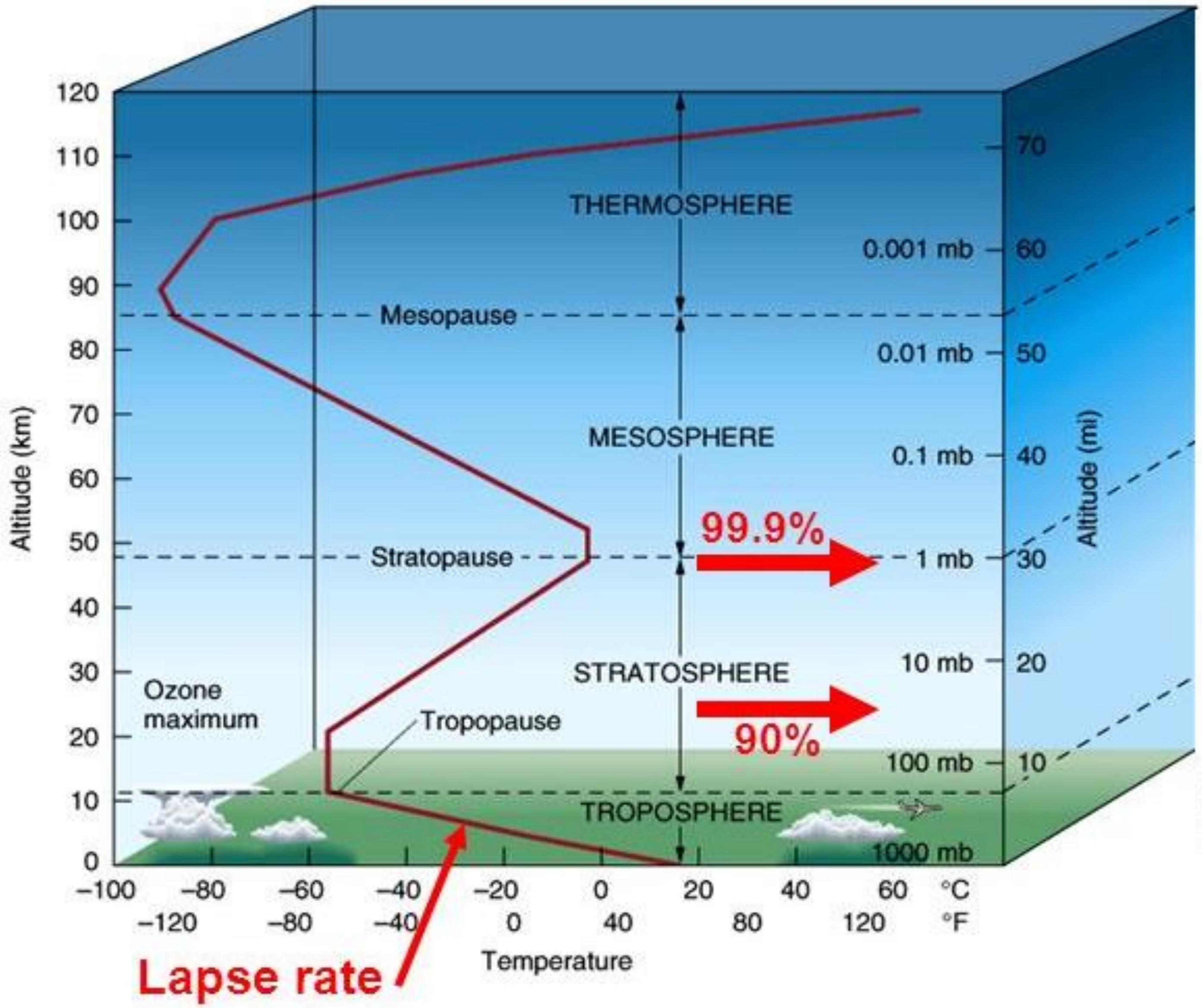
- ❖ Amount of water vapour carried in the air
- ❖ This also decrease rapidly with height and latitude

Decreases half of its surface at 2 km

Less than 10% of its surface value at 5 kms



Layers of the the atmosphere



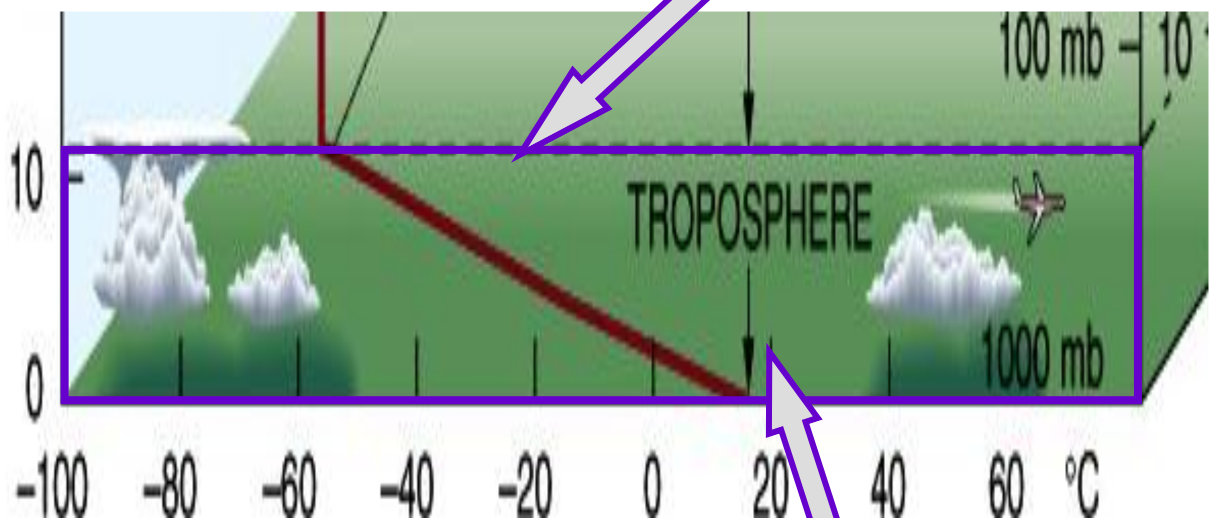
Layer of most interest

Contains 80% of the atmospheric mass

Depth ranges from ~8 km at the poles to ~16 km in the tropics

Troposphere

Tropopause



$$\Gamma \equiv \frac{\partial T}{\partial z} \sim 6.5\ °\mathrm{C\ \ km}^{-1} = 0.0065\ °\mathrm{C\ m}^{-1}$$

The lapse rate is the average decrease in temperature with height ~ 6.5°C/km

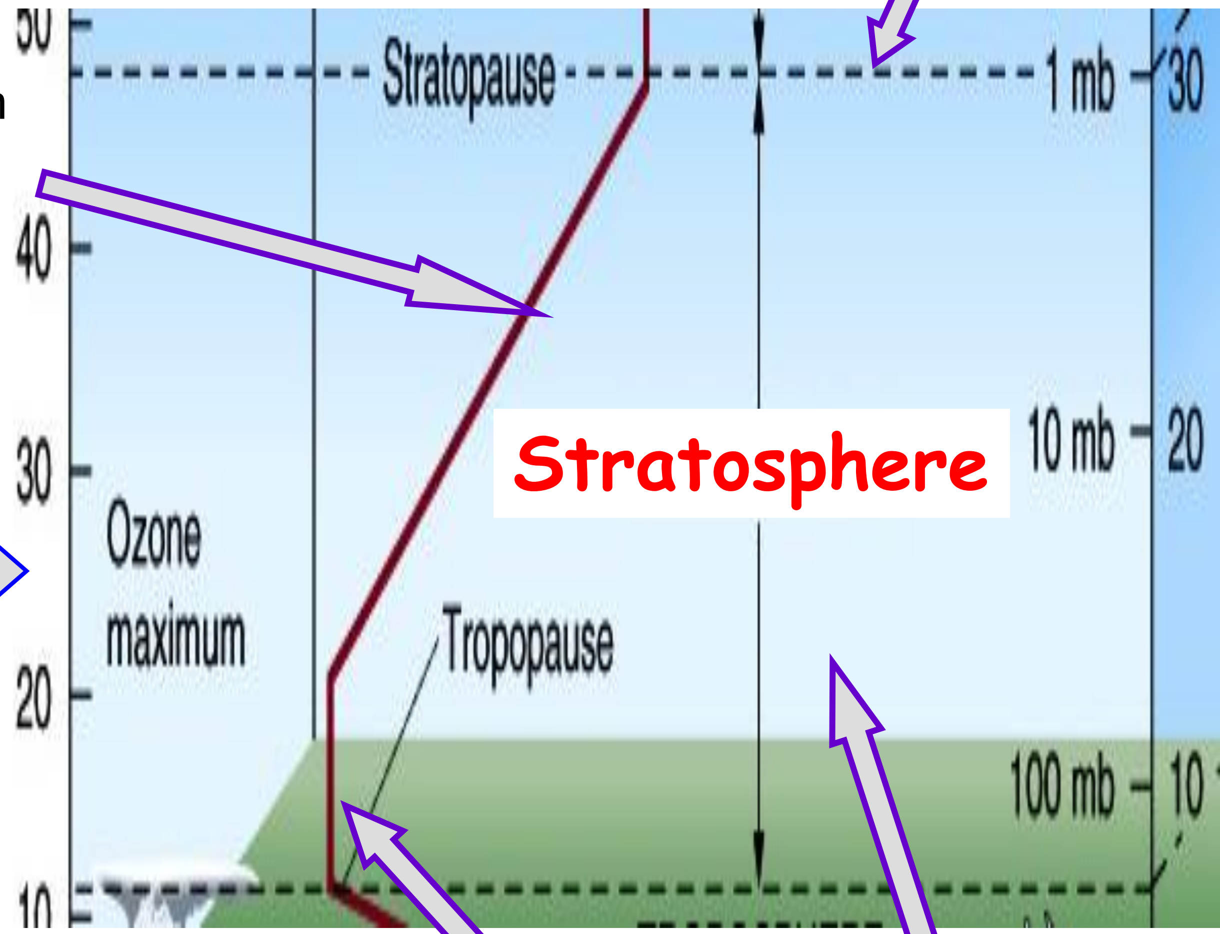
Rapid decrease in temperature with height

Majority of the weather occurs here

Contains ~19.9% of the atmospheric mass

Stratopause

Temperature increases with height from 20-~50 km *(Temperature inversion)*



Ozone layer

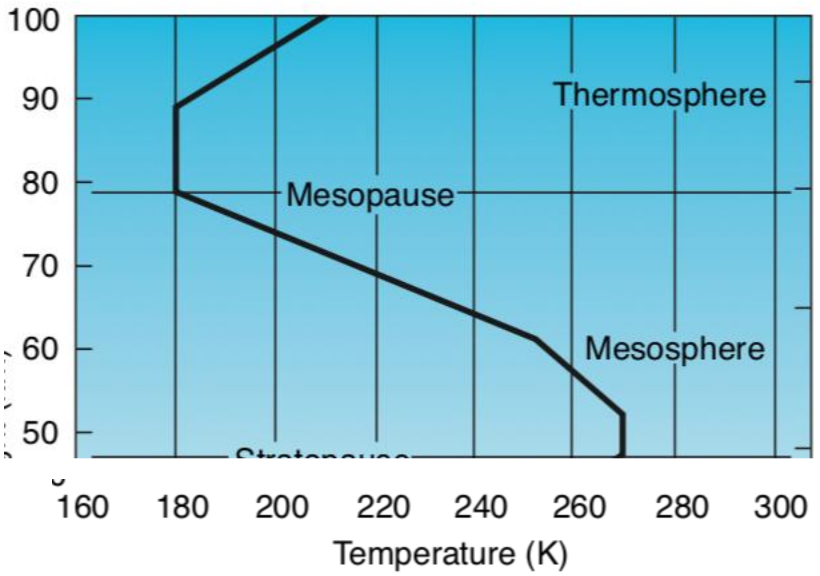
The ozone layer absorbs much of the incoming solar radiation, warming the stratosphere, and protecting us from harmful UV radiation

Isothermal in lowest 10 km
Lapse rate is 0

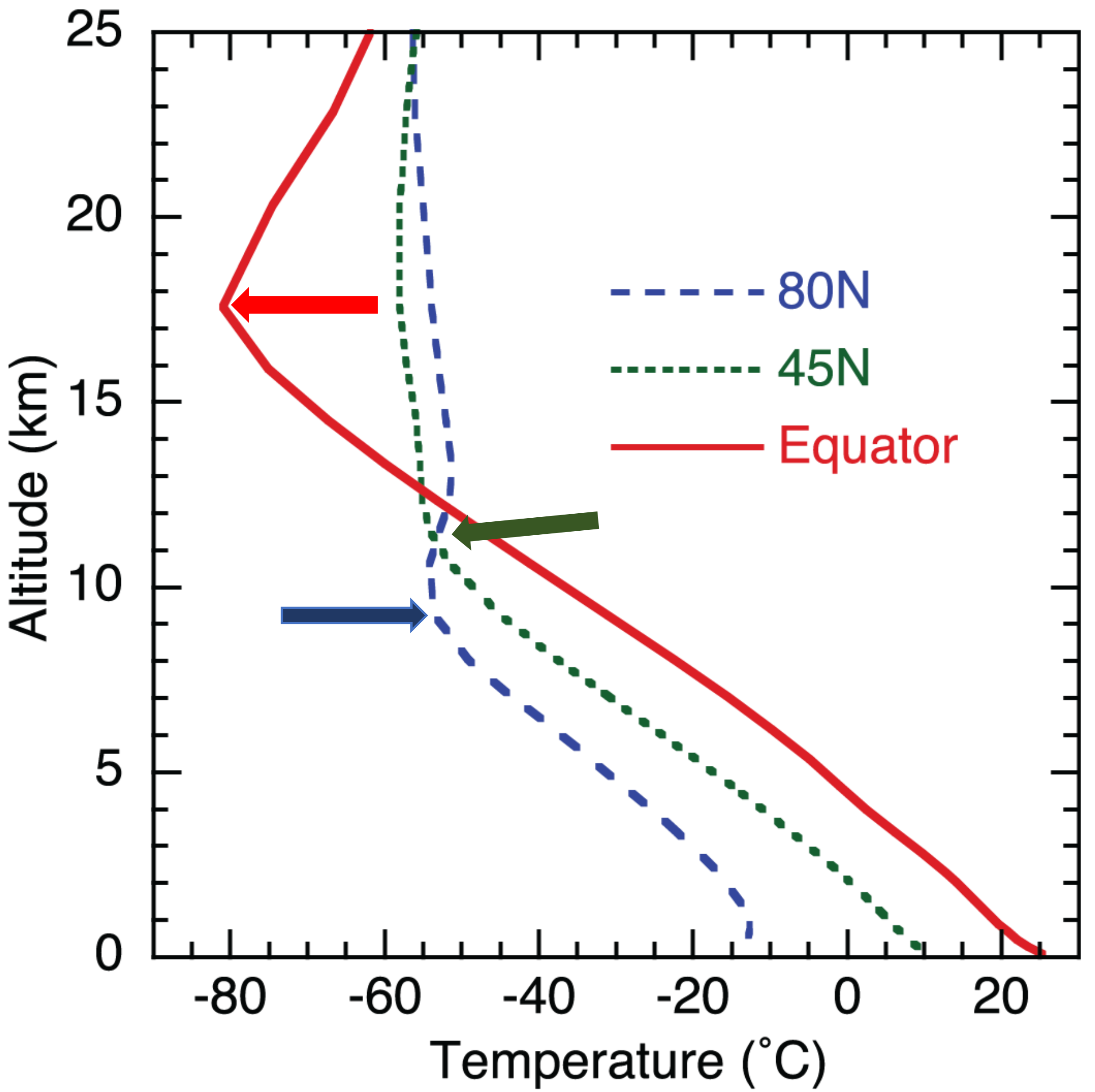
Little weather occurs here

- Stratospheric air is extremely dry and ozone rich.
- The absorption of solar radiation in the ultraviolet region of the spectrum by this stratospheric ozone layer is critical to the habitability of the Earth.
- Heating due to the absorption of ultraviolet radiation by ozone molecules is responsible for the temperature maximum ~50 km that defines the stratopause.

- Above the ozone layer lies the mesosphere (meso connoting "in between"), in which temperature decreases with height to a minimum that defines the mesopause.



- The increase of temperature with height within the thermosphere is due to the absorption of solar radiation in association with the dissociation of diatomic nitrogen and oxygen molecules and the stripping of electrons from atoms.
- These processes, referred to as photodissociation and photoionization



Why is the tropopause so much higher and colder in the Tropics?